

Operator Learning Surrogate Modeling of Hydraulically Fractured Geothermal Injection-Production Systems: A Cornell Case Study

Yulong Liu

Cornell University, Department of Earth and Atmospheric Sciences, New York, Ithaca, USA

Jonah Botvinick-Greenhouse

Cornell University, Center for Applied Mathematics, New York, Ithaca, USA

Yunan Yang

Cornell University, Department of Mathematics, New York, Ithaca, USA

Chloé Arson

Cornell University, Department of Earth and Atmospheric Sciences, New York, Ithaca, USA

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ABSTRACT: Through its Climate Action Plan, Cornell University has committed to achieving carbon neutrality by 2035 and has supported the Earth Source Heat (ESH) project to demonstrate the viability of direct-use deep geothermal energy for district heating on its main campus in Ithaca, NY. To enable rapid inference and optimization of fracture-dominated doublet designs under CUBO site constraints, an operator learning surrogate model is trained on 5,500 finite-element (FE) simulations and evaluated on 498 unseen cases. The surrogate learns the nonlinear solution operator mapping fracture geometry, doublet configuration, and injection mass rate to the temperature field near the production well together with T_{prod} and Q_{net} . It achieves strong predictive accuracy on unseen test cases with field MSE 0.139 ± 0.011 K; T_{prod} MSE 0.503 ± 0.013 K; $R^2 = 0.892$, and evaluates 498 test cases in approximately 0.1 s on a laptop-level device, enabling rapid screening and automatic optimization of well trajectories and spacing under high drilling-cost constraints. Sensitivity and feature importance analyses show that $\dot{m}_{\text{inj}}$ dominantly controls thermal breakthrough and production temperature, while fracture morphology and well spacing act mainly through interaction effects. The effect of fracture scaling s_f is comparatively minor under the present simplified, well-connected fracture setup, and the surrogate reproduces these patterns and sensitivity, indicating that it has learned the underlying FE response behavior.

1 INTRODUCTION

Cornell's energy challenges and decarbonization goals mirror those of New York State and the broader Northeast: approximately 35% of New York's primary energy is used for space heating, accounting for about 30% of the state's greenhouse-gas emissions. In Upstate New York, fossil-fuel-based space heating constitutes roughly 56% of a typical building's total energy use due to the cold winter climate. Comparable reliance on fossil-fuel heating occurs across at least 15 states in the U.S. northern tier (Durga et al., 2021; Mittal, 2022; NYSDEC, 2025). In this context, geothermal district heating provides a scalable, low-carbon pathway to decarbonize campuses and cities

in cold climates. Through its Climate Action Plan, Cornell University has committed to achieving carbon neutrality by 2035 and has supported the Earth Source Heat (ESH) project to demonstrate the viability of direct-use deep geothermal energy for district heating on its main campus in Ithaca, NY (Pinilla et al., 2023; Tester et al., 2023; Purwamaska and Fulton, 2023; Arson et al., 2024; Fulton et al., 2024; Arson et al., 2026).

Preliminary site constraints and information have been established through the Cornell University Borehole Observatory (CUBO), a 10,000-ft borehole drilled in summer 2022 with support from the U.S. Department of Energy (DOE). The primary objective of CUBO is to char-

acterize the subsurface and identify potential fracture-dominated reservoir targets, within both the sedimentary sequence and the basement, where temperatures are expected to fall in the range of 70–90°C. The borehole provides site-specific measurements of in-situ stress, temperature, lithology, hydrogeology, and fracture characteristics, all of which are essential for the design and risk assessment of deep geothermal systems (Fulton et al., 2024). Downhole measurements indicate temperatures of approximately 80°C at 3 km depth, which are sufficiently high for direct-use heating. Incorporating these site-derived constraints can substantially reduce uncertainty in the development of geothermal reservoir models.

The CUBO well penetrated generally low-porosity and low-permeability Paleozoic sedimentary formations before entering metamorphic basement rock near the target reservoir, suggesting that stimulation (e.g., hydraulic fracturing) may be required to create adequate permeability for an enhanced geothermal system (EGS) (Fulcher et al., 2023). Motivated by this site context, the simulations presented here evaluate the thermo-hydraulic response of an injection-production doublet system in a hydraulically fractured reservoir, providing quantitative guidance for well layout and operating strategies during doublet construction.

Designing an injection-production doublet in a fractured reservoir requires resolving strongly coupled thermo-hydro-mechanical (THM) processes governed by the fracture network. Analytical solutions are generally inapplicable, while high-fidelity numerical simulations (e.g., multiphysics finite-element models) remain computationally expensive. This computational burden makes it difficult to solve the coupled Partial Differential Equations (PDEs) repeatedly for design iterations, and to optimize lateral well trajectories and well spacing in real time. Consequently, identifying designs that maximize heat extraction while limiting thermal drawdown and delaying thermal breakthrough remains a persistent challenge.

To address these limitations, and motivated by recent advances in gradient-based machine learning surrogates for solving PDE governed systems (Khorrami et al., 2023; Liu and Arson, 2025, 2026; Feng et al., 2025, 2026), we aim to develop an operator learning surrogate that maps boundary conditions, well controls, doublet placement, and fracture network geometry to key thermo-hydraulic (TH) responses (e.g., temperature fields and produced thermal power). The surrogate will be trained and conditioned on the site-specific geological, hydrogeological, and thermal constraints provided by CUBO. Our central hypothesis is that such a surrogate can learn the nonlinear solution oper-

ator that maps fracture-network structure, operating controls, and doublet configuration to fracture-controlled TH responses, thereby enabling rapid optimization of lateral well trajectories and placements during doublet design.

2 METHODOLOGY

2.1. Numerical simulation

2.1.1 Model assumptions

Thermal and hydraulic properties measured at CUBO are used to characterize the geothermal reservoir at Cornell University’s main campus. In anticipation of the construction of an EGS, the reservoir is modeled as rock formation of low porosity intersected by a discrete fracture network including a hydraulically-induced fracture which connects an injection well to a production well. Coupled thermo-hydraulic (TH) transport equations are solved for the pore pressure and temperature fields during the operation of the doublet. Fractures are treated as a porous medium distinct from the matrix and are discretized using dedicated fracture elements in the finite-element mesh, yielding a dual-porosity representation. The mathematical model is established under the following assumptions:

1. The fractures are embedded in an isotropic, homogeneous, and fully saturated rock matrix.
2. The pore fluid is single-phase water and its flow obeys Darcy’s law; capillary effects and phase change are neglected.
3. Heat transfer is governed by Fourier’s law. Local thermal equilibrium is assumed between the solid and fluid phases, and heat transport occurs via advection in the flowing fluid and conduction through the porous medium.
4. The thermal and hydraulic properties of the matrix and fractures are constant throughout the simulations.

2.1.2 Governing equations

Mass conservation for the single-phase pore fluid is described by the continuity equation

$$\frac{\partial M}{\partial t} + \nabla \cdot \mathbf{F} = q, \quad (1)$$

where M is the fluid mass per bulk volume ($\text{kg} \cdot \text{m}^{-3}$), $\mathbf{F}$ is the fluid mass flux vector ($\text{kg} \cdot \text{m}^{-2} \cdot \text{s}^{-1}$), and q is a prescribed mass source term ($\text{kg} \cdot \text{m}^{-3} \cdot \text{s}^{-1}$). Mechanical coupling, chemical reactions, and diffusive/dispersion mass fluxes are neglected.

For a fully saturated, single-phase system,

$$M = \phi \rho, \quad (2)$$

where ϕ is porosity and ρ is the fluid density ($\text{kg} \cdot \text{m}^{-3}$).

With advection only (no diffusion/dispersion),

$$\mathbf{F} = \rho \mathbf{v}, \quad (3)$$

where $\mathbf{v}$ is the Darcy velocity ($\text{m} \cdot \text{s}^{-1}$). The Darcy flux obeys Darcy's law,

$$\mathbf{v} = -\frac{\mathbf{k}}{\mu} (\nabla p - \rho \mathbf{g}), \quad (4)$$

where $\mathbf{k}$ is the absolute permeability tensor (m^2), μ is the viscosity ($\text{Pa} \cdot \text{s}$), p is pore pressure (Pa), and $\mathbf{g}$ is the gravitational acceleration vector ($\text{m} \cdot \text{s}^{-2}$).

Energy conservation for heat transport is written in the same continuity form as

$$0 = \frac{\partial \mathcal{E}}{\partial t} + \nabla \cdot \mathbf{F}^\top - q^\top, \quad (5)$$

where $\mathcal{E}$ is the volumetric heat energy density of the rock-fluid system ($\text{J} \cdot \text{m}^{-3}$), $\mathbf{F}^\top$ is the total heat flux ($\text{W} \cdot \text{m}^{-2}$), and $q^\top$ is a volumetric heat source term ($\text{W} \cdot \text{m}^{-3}$). Mechanical coupling terms and chemical reaction terms are neglected, and local thermal equilibrium (LTE) is assumed so that a single temperature T characterizes both the solid and fluid phases.

Under the assumption of LTE for a fully saturated, single-phase system, the energy density is

$$\mathcal{E} = (1 - \phi) \rho_s c_s T + \phi \rho c_p T, \quad (6)$$

where ϕ is porosity, ρ_s and c_s are the density ($\text{kg} \cdot \text{m}^{-3}$) and specific heat capacity ($\text{J} \cdot \text{kg}^{-1} \cdot \text{K}^{-1}$) of the solid phase (when considered non-porous), and ρ and c_p are the density ($\text{kg} \cdot \text{m}^{-3}$) and specific heat capacity ($\text{J} \cdot \text{kg}^{-1} \cdot \text{K}^{-1}$) of the fluid.

The heat flux is the sum of conductive and advective contributions (no thermal dispersion),

$$\mathbf{F}^\top = -\lambda \nabla T + h \mathbf{F}, \quad (7)$$

where λ is the effective thermal conductivity of the porous medium ($\text{W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$), h is the fluid specific enthalpy ($\text{J} \cdot \text{kg}^{-1}$), and $\mathbf{F}$ is the advective Darcy flux ($\text{kg} \cdot \text{m}^{-2} \cdot \text{s}^{-1}$) defined in Eq. (3).

2.1.3 Benchmark with analytical solution

Before proceeding to large-scale data generation, we benchmark the model presented in Subsection 2.1.2 against the analytical solution of Lauwerier (1955), which describes heat transport in a single fracture embedded in an impermeable rock matrix. In the corresponding

2D single-fracture idealization, both the fracture and the surrounding matrix extend to infinity in the x - and z -directions. The system is initially at a uniform temperature T_0 . Water is injected into the fracture at a constant temperature $T_{in} < T_0$ and with a constant velocity u_{in} , thereby extracting heat from the matrix. The analytical solution yields the following expressions.

$$\chi(x, t) \equiv \frac{\lambda_s x / (\rho_f c_{p,f} d_f)}{\sqrt{u_{in}(u_{in} t - x) \lambda_s / (\rho_s c_{p,s})}}, \quad (8)$$

$$T_f(x, t) = T_i + (T_{in} - T_i) \operatorname{erfc}(\chi(x, t)) U\left(t - \frac{x}{u_{in}}\right), \quad (9)$$

where T_f (K) is the fluid temperature along the fracture, ρ (kg/m^3) refers to density, c_p ($\text{J}/(\text{kg} \cdot \text{K})$) refers to specific heat capacity, and λ ($\text{W}/(\text{m} \cdot \text{K})$) is the thermal conductivity. The subscripts s and f denote the rock matrix (solid) and the fluid, respectively. In Eq. (9), $\operatorname{erfc}(\cdot)$ is the complementary error function and $U(\cdot)$ is the unit step function. The parameters used in this analytical benchmark are summarized in Table. 1.

Table 1: Parameters used for Lauwerier's analytical solution.

Category	Symbol	Value	Unit
Temperatures	T_0	80	$^\circ\text{C}$
	T_{in}	30	$^\circ\text{C}$
Rock properties	ρ_s	2700	kg m^{-3}
	λ_s	3	$\text{W m}^{-1} \text{K}^{-1}$
	$c_{p,s}$	1000	$\text{J kg}^{-1} \text{K}^{-1}$
Fluid (water) properties	ρ_f	1000	kg m^{-3}
	$c_{p,f}$	4200	$\text{J kg}^{-1} \text{K}^{-1}$
Fracture / flow	u_f	0.02	m s^{-1}
	d_f	8×10^{-4}	m

To benchmark the model presented in Subsection 2.1.2 against the analytical solution, finite-element simulations are performed in the MOOSE framework (Permann et al., 2020) using a MultiApp setup, in which the governing TH physics are solved in separate, coupled sub-applications rather than within a single monolithic fully coupled solver. The rock matrix is modeled as a purely thermal domain, where temperature evolves by conduction and exchanges heat with the fracture fluid. Mass and heat transport in the fracture are modeled as non-isothermal single-phase fluid flow in a fully saturated porous medium. The parameters used in the MOOSE simulation are listed in Table. 2. The matrix domain is a two-dimensional $100\text{m} \times 100\text{m}$ region representing the rock mass, within which a horizontal fracture of aperture d_f and length 100m is embedded at mid-depth.

Table 2: Parameters used in the MOOSE MultiApp benchmark.

Category	Symbol	Value	Unit
<i>Geometry / mesh</i>			
Fracture length	L_f	100	m
<i>Temperatures</i>			
Initial temperature	T_0	353.15	K
Injection temperature	T_{in}	303.15	K
<i>Matrix thermal properties</i>			
Density	ρ_s	2700	kg m^{-3}
Specific heat capacity	$c_{p,s}$	1000	$\text{J kg}^{-1} \text{K}^{-1}$
Thermal conductivity	λ_s	3	$\text{W m}^{-1} \text{K}^{-1}$
<i>Fluid properties</i>			
Reference density	$\rho_{f,0}$	1000	kg m^{-3}
Heat capacity	$c_{p,f}$	4200	$\text{J kg}^{-1} \text{K}^{-1}$
Viscosity	μ_f	10^{-3}	Pas
Bulk modulus	K_f	2×10^{20}	Pa
Thermal expansion	β_T	0	K^{-1}
<i>Fracture hydraulic / thermal properties</i>			
Aperture	d_f	8×10^{-4}	m
porosity	a	8×10^{-4}	m
Permeability (isotropic)	k_f	4.27×10^{-11}	m^2
Dry thermal conductivity	$\lambda_{f,dry}$	1.6×10^{-30}	$\text{W m}^{-1} \text{K}^{-1}$
<i>Initial and boundary conditions</i>			
Initial pressure	$P_{f,0}$	2.0×10^7	Pa
Injection mass rate	$\dot{m}_{inj}$	1.6×10^{-2}	kg s^{-1}
injection and production well pressure difference	P_{bh}	10^2	Pa

The analytical and numerical results are compared in Fig. 1. The two solutions show excellent agreement, validating the MOOSE PorousFlow implementation of flow and heat transport in a single lower-dimension fracture.

2.1.4 Numerical implementation

Two-dimensional thermo-hydraulic transport in fracture networks is simulated with the Finite Element Method (FEM), in MOOSE. We consider an idealized enhanced geothermal system doublet in which the rock matrix is homogeneous, isotropic, and saturated with water. The matrix is represented by a two-dimensional $1500\text{m} \times 1500\text{m}$ domain intersected by a lower-dimensional (1D) fracture network. Both the injection and production wells are located within the interior of the domain; the injection well is positioned slightly below the production well, and the fracture network lies in the region between the two wells. Cold water is injected through the lower injection well with a prescribed constant mass injection rate $\dot{M}_{inj}$. The upper production well is operated under a fixed bottom-hole pressure p_{prod} . All simulations are performed over a three year injection-production period.

Initial conditions are prescribed to represent the undisturbed hydrostatic and geothermal state. The initial pore pressure is calculated based on depth, with a reference pressure of 32 MPa at mid-depth, and a linear hydrostatic gradient of $9.81 \text{ kPa} \cdot \text{m}^{-1}$. The initial temperature is constrained by CUBO well-logging data, and is set to 354.15 K at mid-depth, with a linear thermal gradient of

$0.023 \text{ K} \cdot \text{m}^{-1}$. Dirichlet boundary conditions for both pressure and temperature are applied on all outer boundaries and set equal to these initial-state values, representing a far-field environment that remains at undisturbed conditions. The model configuration and boundary conditions are illustrated in Fig. 2.

Matrix properties are taken as representative averages inferred from CUBO well logging data (Beckers et al., 2024). The matrix porosity is set to $\phi_m = 3.9\%$, and the stimulated reservoir is assigned an intrinsic permeability of $k_m = 5 \times 10^{-15} \text{ m}^2$. The density of the solid phase is $\rho_{s,m} = 2800 \text{ kg} \cdot \text{m}^{-3}$, the specific heat capacity of the solid phase is $c_{s,m} = 1000 \text{ J} \cdot \text{kg}^{-1} \cdot \text{K}^{-1}$, and the matrix thermal conductivity is $\lambda_m = 2.8 \text{ W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$.

Hydraulic fractures are represented as a high-permeability porous continuum with aperture $d_f = 6 \times 10^{-4} \text{ m}$ and porosity $\phi_f = 1$. The fracture permeability is set to $k_f = 3 \times 10^{-8} \text{ m}^2$, the solid density to $\rho_{s,f} = 0 \text{ kg} \cdot \text{m}^{-3}$, the specific heat capacity to $c_{s,f} = 800 \text{ J} \cdot \text{kg}^{-1} \cdot \text{K}^{-1}$, and the thermal conductivity to $\lambda_f = 2.0 \text{ W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$.

For the fluid, we assume single-phase flow governed by Darcy’s law. The fluid bulk modulus is $K_f = 1 \text{ GPa}$, the reference density is $\rho_0 = 1000 \text{ kg} \cdot \text{m}^{-3}$, the viscosity is $\mu = 1.0 \times 10^{-3} \text{ Pa} \cdot \text{s}$, and the specific heat capacity is $c_{p,f} = 4186 \text{ J} \cdot \text{kg}^{-1} \cdot \text{K}^{-1}$. Fluid density is modeled as

$$\rho = \rho_0 \exp\left(\frac{p}{K_f} - \alpha_f T\right), \quad (10)$$

where p is pore pressure (Pa), T is temperature (K), and α_f is the volumetric thermal expansion coefficient of the fluid (K^{-1}).

Each simulation in the dataset uses a prescribed set of operational parameters consistent with the ranges summarized in Table 3. The injection mass rate is selected from the discrete set $\{20.0, 30.0, 40.0, 50.0\} \text{ kg} \cdot \text{s}^{-1}$, while the injection temperature and production pressure are held constant at 300 K and 20 MPa, respectively.

The injection-well location is fixed at $(0, -250)$, while the production-well location varies across simulations only in the vertical direction, with $x = 0$ and $y \in [0, 200]$. The fracture network comprises eight hydraulic fractures, including two vertical fractures with fixed locations that emulate hydraulically-induced fractures branching from the wells. To support the Cornell case study while controlling the dimensionality of the input space, the remaining six fractures share a fixed topology and are parameterized only by (i) a rotation angle (continuous, $0 \sim 90^\circ$, clockwise or counterclockwise) and (ii) a dilation factor selected from the discrete set $[0.6, 0.8, 1.0, 1.2, 1.4]$. This

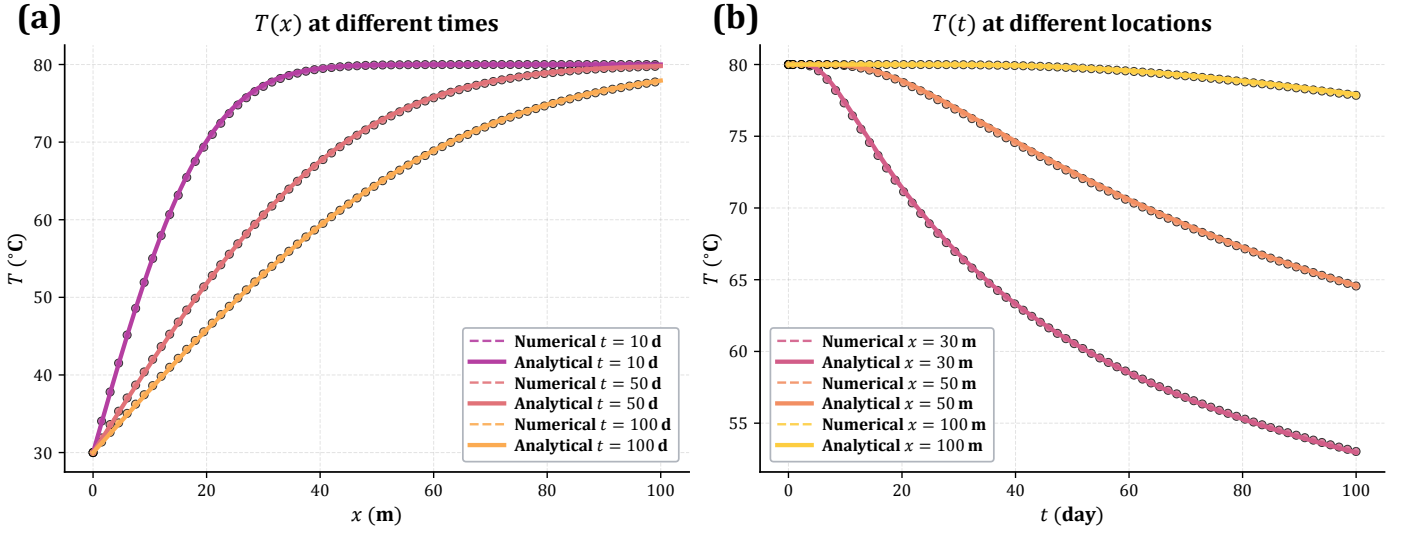


Fig. 1: Benchmark of the MOOSE simulation against Lauwerier's analytical solution for heat transport in a single fracture. (a) Temperature evolution $T(x,t)$ along the fracture at $t = 10, 50$, and 100 days. (b) Temperature evolution $T(t)$ at $x = 30, 50$, and 100 m. Solid lines denote the analytical solution and markers denote the numerical results.

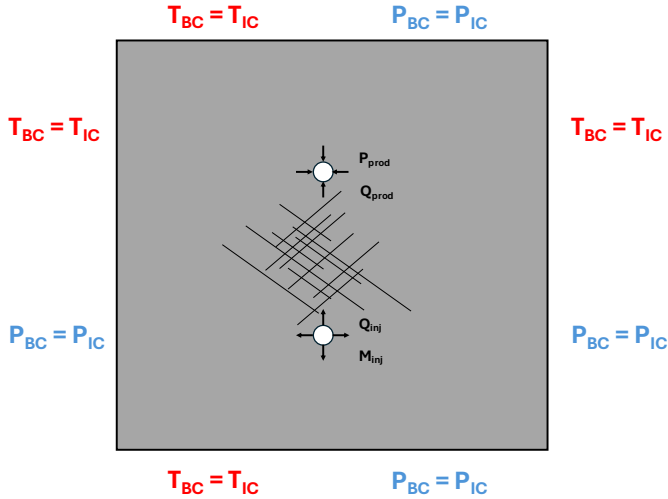


Fig. 2: Illustration of the boundary conditions for a doublet enhanced geothermal system simulated in a fractured reservoir. Cold water is injected through the lower injection well with a prescribed constant mass injection rate M_{inj} . The upper production well is operated under a fixed bottom-hole pressure p_{prod} . The top, bottom, left, and right outer boundaries impose far-field pressure and temperature conditions with prescribed linear gradients; boundary values are consistent with the initial conditions to represent an undisturbed geothermal gradient and far-field hydraulic state.

parameterization substantially reduces the effective input dimension. The mesh is locally refined around the fracture network and in the vicinity of the injection and production wells.

2.1.5 EGS performance evaluation parameters

The metrics used to evaluate EGS performance in the simulations include the production temperature and the output thermal power (Lei et al., 2023). The production temperature at time t is denoted by $T_{pro}(t)$. The output thermal power Q_{net} represents the net heat production rate of the EGS and is computed as

$$Q_{net} = \dot{m}_{inj} C_w [T_{pro}(t) - T_{inj}], \quad (11)$$

where $\dot{m}_{inj}$ (kg s^{-1}) is the injection mass flow rate, C_w ($\text{J kg}^{-1} \text{K}^{-1}$) is the specific heat capacity of the working fluid, T_{inj} is the injection temperature, and $T_{pro}(t)$ is the production temperature.

The production temperature is a primary indicator of delivered heat quality; tracking $T_{pro}(t)$ ensures that the extracted hot water remains sufficiently warm to meet the temperature requirement of direct-use district heating for campus demand. In parallel, the output thermal power Q_{net} indicates how much heating the geothermal system can deliver, and thus how effectively it can replace fossil fuel based space heating and meet the required thermal demand.

2.2. Operator learning

While traditional deep learning methods approximate non-linear functions between finite-dimensional spaces, operator learning (OL) provides a framework for learning

Table 3: Range of parameters used in reservoir simulations.

Name	Symbol	Value / Range
<i>Geometry / configuration</i>		
Injection well location	$(x_{\text{inj}}, y_{\text{inj}})$	$(0, -250, 0)$
Production location	$(x_{\text{prod}}, y_{\text{prod}})$	$y_{\text{prod}} \in [0, 200]$ m
<i>Initial conditions</i>		
Initial pore pressure	$p_{p,0}$	32 MPa with gradient 9.81 kPa/m
Initial temperature	T_0	354.15 K with gradient 0.023 K/m
<i>Operational controls</i>		
Injection mass rate	$\dot{m}_{\text{inj}}$	{20.0, 30.0, 40.0, 50.0} kg · s ⁻¹
Injection temperature	T_{inj}	300 K
Production pressure	p_{prod}	20 MPa
<i>Fluid properties (constant)</i>		
Fluid bulk modulus	K_f	1 GPa
Reference fluid density	$\rho_{f,0}$	1000 kg/m ³
Fluid thermal expansion	α_f	0
Fluid viscosity	μ_f	0.001 Pa · s
<i>Matrix properties (constant)</i>		
Matrix porosity	ϕ_m	3.9%
Matrix permeability	k_m	5×10^{-15} m ²
Matrix solid density	$\rho_{s,m}$	2800 kg/m ³
Matrix heat capacity	$c_{p,s,m}$	1000 J/(kg · K)
Matrix thermal conductivity	$k_{th,m}$	2.8 W/(m · K)
<i>Fracture properties</i>		
Fracture porosity	ϕ_f	1
Fracture permeability	k_f	3×10^{-8} m ²
Fracture solid density	$\rho_{s,f}$	0 kg/m ³
Fracture heat capacity	$c_{p,s,f}$	800 J/(kg · K)
Thermal conductivity	$k_{th,f}$	2 W/(m · K)
Fracture rotation angle	θ_f	[0, 180°]
Fracture scaling factor	s_f	{0.6, 0.8, 1.0, 1.2, 1.4}

maps between infinite-dimensional spaces of functions (Kovachki et al. (2024)). It has been successfully deployed to approximate solution operators for a wide range of PDEs appearing across fluid mechanics, geophysics, and biology (You et al. (2022); Wang et al. (2024); Diab and Al Kobaisi (2024)). We propose to use OL to develop an efficient surrogate model for the FEM simulation, in which the fracture network geometry, wellbores layout and operational parameters are used to predict the temperature field in the vicinity of the production well.

2.2.1 Data generation and preprocessing

Finite-element simulations are used to generate the training and testing datasets for the OL framework. A Python script is developed to generate $N = 6,000$ simulation cases for OL. In each case, operational controls and fracture

network parameters are randomized by sampling from the ranges specified in Table 3 to promote generalizability. Production metrics, including the production temperature and produced thermal power, are extracted at the production-well location and post-processed in MOOSE, providing scalar labels in addition to the full-field ground truth. To prepare the data for OL, all temperature fields are linearly interpolated to a common uniform mesh, and all functions and operational parameters are normalized by affine transformations to take values in $[-1, 1]$.

2.2.2 Model setup

We write ψ to denote the fracture network function

$$\psi(\mathbf{x}) = \begin{cases} 1 & \mathbf{x} \in F \\ 0 & \mathbf{x} \notin F \end{cases},$$

where F is the subset of the domain where fractures occur. Given the fracture network function and operational parameters y_{prod} and $\dot{m}_{\text{inj}}$, our goal is to predict the temperature field in a neighborhood of the production well after 3 years of time evolution. In particular, we focus on predicting $T|_{\Omega(\mathbf{x}_{\text{prod}})}$ with $\Omega(\mathbf{x}_{\text{prod}}) = \mathbf{x}_{\text{prod}} + \mathbf{R}$ where $\mathbf{R} \subseteq \mathbb{R}^2$ is a small rectangle centered at the origin. A visualization of $\Omega(\mathbf{x}_{\text{prod}})$ is shown in the top row of Figure 3. Thus, we seek to learn the operator

$$\mathcal{G}[\psi, \mathbf{x}_{\text{prod}}, \dot{m}_{\text{inj}}] = T|_{\Omega(\mathbf{x}_{\text{prod}})}. \quad (12)$$

To reduce the computational cost of simulating (12) using the FEM, we approximate $\mathcal{G}$ using neural networks. To promote stable gradient-based training and avoid discontinuous inputs, we replace the binary fracture indicator ψ with a smoothed distance-based encoding:

$$\psi_\varepsilon(\mathbf{x}) = \exp(-D(\mathbf{x})/\varepsilon), \quad D(\mathbf{x}) = \min_{\mathbf{y} \in \{\psi=1\}} \|\mathbf{x} - \mathbf{y}\|. \quad (13)$$

In (13), $\|\cdot\|$ denotes the Euclidean distance and $\varepsilon > 0$ is a constant determining the smoothness of ψ_ε ; a visualization of ψ_ε is shown in the top row of Figure 3. Writing $u = (\psi_\varepsilon, y_{\text{prod}}, \dot{m}_{\text{inj}})$ and $v = T|_{\Omega(\mathbf{x}_{\text{prod}})}$, we learn the mapping $u \mapsto v$ in place of (12).

We approximate (12) using the DeepONet architecture (Lu et al. (2021)), given by

$$\mathcal{G}_\theta[u](\mathbf{z}) = \sum_{k=1}^p b_k(u; \theta) r_k(\mathbf{z}; \theta). \quad (14)$$

In (14), $\mathbf{z} = \mathbf{x} - \mathbf{x}_{\text{prod}} \in \mathbf{R}$ parameterizes the local coordinates centered at the production well, $\theta \in \mathbb{R}^\ell$ comprises the model's tunable parameters and $p \in \mathbb{Z}_+$ is a hyperparameter constraining the rank of $\mathcal{G}_\theta$. The trunk network r takes

spatial coordinates as inputs and produces p basis functions, while the branch network b evaluates the fracture network $\psi_\varepsilon = [\psi_\varepsilon(\mathbf{x}_1) \dots \psi_\varepsilon(\mathbf{x}_{4096})]$ discretized on a 64×64 grid and the parameters $(y_{\text{prod}}, \dot{m}_{\text{inj}})$ to determine weights for these basis functions.

The trunk network $r : \mathbf{R} \rightarrow \mathbb{R}^p$ is constructed as a standard feed-forward neural network with ReLU activation. The branch network is represented by

$$b(u) = h(f(\psi_\varepsilon), g(y_{\text{prod}}, \dot{m}_{\text{inj}})),$$

where $f : \mathbb{R}^{4096} \rightarrow \mathbb{R}^m$ encodes a latent m -dimensional representation of the fracture field and production location, $g : \mathbb{R}^2 \rightarrow \mathbb{R}^m$ encodes a latent representation of the operational parameters, and $h : \mathbb{R}^{2m} \rightarrow \mathbb{R}^p$ uses these latent representations to predict the basis coefficients. The networks f, g , and h are all parameterized as feed forward networks with ReLU activation functions.

The FEM simulations yield a dataset of high-fidelity inputs $\{u_i\}_{i=1}^N$ and outputs $\{v_i\}_{i=1}^N$ which are used to train the model (14) by minimizing the mean-squared error loss

$$\mathcal{J}(\theta) = \frac{1}{KM} \sum_{j=1}^K \sum_{n=1}^M (\mathcal{G}_\theta[u_j](\mathbf{z}_n) - v_j(\mathbf{z}_n))^2. \quad (15)$$

We use the Adam optimizer (Kingma and Ba (2015)) to minimize (15), where minibatches of K functions $\{(u_j, v_j)\}_{j=1}^K$ and M sample points $\{\mathbf{z}_n\}_{n=1}^M$ are drawn with replacement each step of the optimization.

3 NUMERICAL RESULTS

We now present results for training the OL on input-output data from high-fidelity FEM simulations. The training set consists of $5.5 \cdot 10^3$ samples while 498 samples are reserved for testing. We construct the DeepONet (14) architecture using $p = 50$ basis functions. DeepONet is selected because it separates geometric input encoding from spatial basis representation, aligning naturally with the operator structure of parametric PDE solutions. The trunk network r has four hidden layers with 200 neurons each, the embedding layers f and g of the branch network both have one hidden layer with 50 neurons each and map to a latent dimension of $m = 5$. Finally, the network h has four hidden layers with 200 neurons each. In total, the OL architecture has approximately $\ell \approx 5 \cdot 10^5$ tunable parameters. We minimize the objective (15) using a learning rate of 10^{-3} and a total of 10^5 optimization steps. At each step, we evaluate (15) using minibatches of $K = 10^2$ functions and $M = 10^3$ sample points.

The trained OL surrogate achieves high accuracy in predicting both the production temperature and the temper-

ature field in the vicinity of the production window. Although minor localized artifacts are present in some predictions, Fig. 3 demonstrates that the model captures both short-circuiting behavior (first column) and well-heated cases (second column), as well as fracture-controlled flow patterns and conductive heat exchange with the surrounding matrix. While the FEM solution is costly to compute, the OL surrogate model can be evaluated rapidly on all 498 testing examples in approximately 0.1 seconds using laptop-level computing.

To highlight reproducibility and consistency of the OL model, we trained the neural network 10 times with different randomized initial conditions. We found that all initializations converge to low-error solutions of the model with low variance in generalization error. In particular, we computed the mean-squared error (MSE) between the OL prediction and the FEM solution for all examples in the testing set and obtained 0.139 ± 0.011 (K). We next used the OL surrogate to extract the temperature at the production well. We then compared this value to the ground truth production temperature from the FEM model across the 10 training instances, resulting in a mean squared error of 0.503 ± 0.013 (K). Figure 4 shows a linear regression between the mean OL production temperature and the FEM solution across all testing examples, yielding a slope of 1.114 and an R^2 value of 0.892. Standard deviation of uncertainty is also shown for each prediction in Figure 4.

4 SENSITIVITY ANALYSIS AND DISCUSSION

4.1. Univariate sensitivity analysis

We perform univariate sensitivity analysis to assess how the input parameters affect the two output metrics, and to verify whether the model has learned the underlying thermo-hydraulic operator. For the well position parameter, the FEM results in Fig.5(a) show that the mean production temperature T_{prod} depends only weakly on y_{prod} . However, the 10th-90th percentile band expands at smaller y_{prod} (shorter injection and production well spacing), indicating that T_{prod} becomes more sensitive on the remaining inputs and that certain parameter combinations can induce substantially larger variability under close-well configurations. Over the same range, the mean net thermal power Q_{net} exhibits only a mild dependence on y_{prod} (Fig.5(c)). The OL predictions (Fig. 5(b,d)) closely reproduce both the mean trends and the spread of the FEM responses, indicating that the surrogate reproduces the dominant FE response trends.

A similar interpretation applies to the fracture rotation parameter. As shown in Fig. 6(a,c), the fracture rotation angle θ_f produces a limited first-order sensitivity on

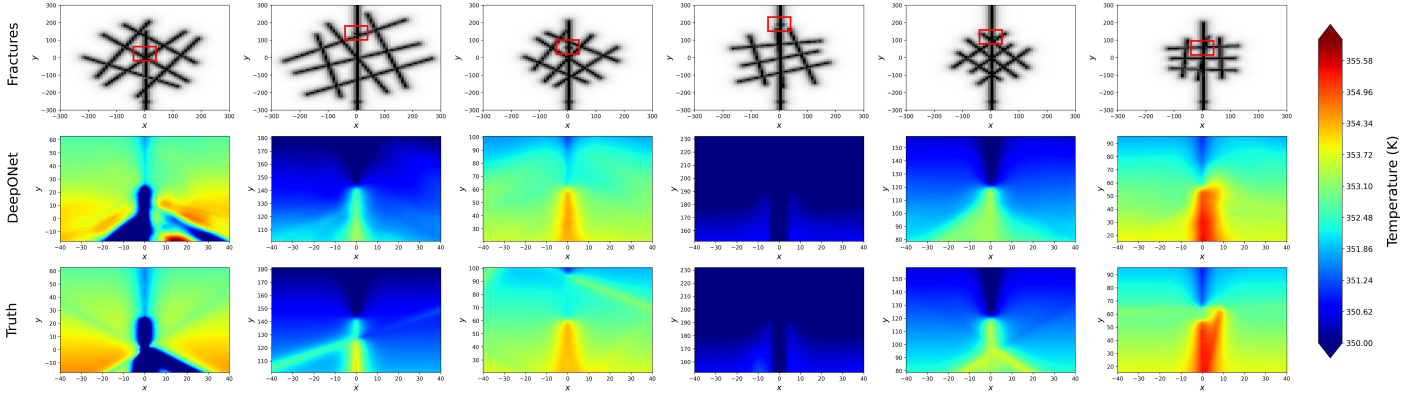


Fig. 3: Generalization of the operator learning on six unseen testing examples. The regularized input fracture networks ψ_ε and rectangular subdomain $\Omega(\mathbf{x}_{\text{prod}})$ are shown in the top row. The output of the operator learning evaluated on the fracture network and operational parameters is shown in the middle row, while the ground truth FEM solution is plotted in the bottom row.

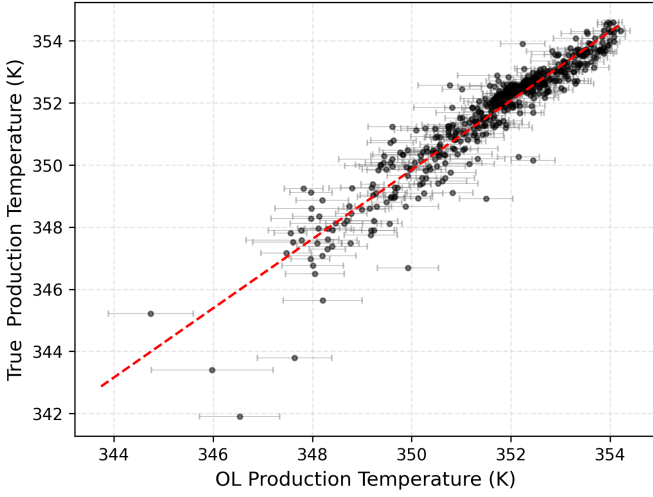


Fig. 4: Linear regression between the OL prediction for production temperature and the ground truth FEM solution. Predictions are averaged over 10 instances of neural network training with one standard deviation of uncertainty visualized.

the mean values of T_{prod} and Q_{net} , whereas the percentile bands remain broad and vary with θ_f . In particular, the widening of the T_{prod} band at specific angles indicates that fracture morphology primarily influences the outputs through coupling with other inputs, e.g., $(y_{\text{prod}}, s_f, \dot{m}_{\text{inj}})$, rather than as an isolated driver. Certain angles may also increase the likelihood of flow short-circuiting, which can lead to markedly lower production temperatures. The OL surrogate reproduces these mean and variability patterns, further confirming that it has learned the sensitivity present in the FEM data.

Fig. 7 shows that increasing the injection rate promotes stronger cold-water invasion and earlier thermal breakthrough, thereby reducing the production temperature, consistent with physical intuition. Meanwhile, a higher

injection rate can increase the heat-extraction power: even if T_{prod} decreases, the larger mass flow rate may dominate the net-power calculation, meaning the gain from increased flow outweighs the loss from a reduced temperature difference, leading to an overall increase in W_h . The OL surrogate captures the same trend as shown in Fig. 7(b,d).

Fig. 8 shows that the boxplots for different s_f do not exhibit a systematic upward or downward shift, indicating that fracture scaling alone does not produce a monotonic change in the outputs. Instead, its impact emerges primarily through interactions with other inputs. The OL predictions show minor discrepancies in some cases, but the overall trends remain consistent with the FEM results.

Overall, except for $\dot{m}_{\text{inj}}$, which has a clear and direct influence on the responses, most inputs affect the outcomes mainly through coupling effects. This further motivates the use of an OL surrogate to efficiently evaluate coupled combinations of well placement and operating controls, and to identify well layouts that promote favorable fracture flow pathways, thereby balancing: (i) sufficient sweep and heat-exchange area, (ii) avoidance of excessive separation that reduces net thermal output, and (iii) favorable fracture-controlled flow paths that delay thermal breakthrough.

4.2. Feature importance ranking

Feature importance is assessed using random forest mean decrease in impurity (Gini importance). We note that Gini importance may be biased toward features with larger variance; results are interpreted qualitatively. For each target metric, a RandomForestRegressor is trained using the four input features $(y_{\text{prod}}, \theta_f, s_f, \dot{m}_{\text{inj}})$, and the normalized impurity-reduction contribution of each feature is computed. To ensure a fair comparison between the FEM and

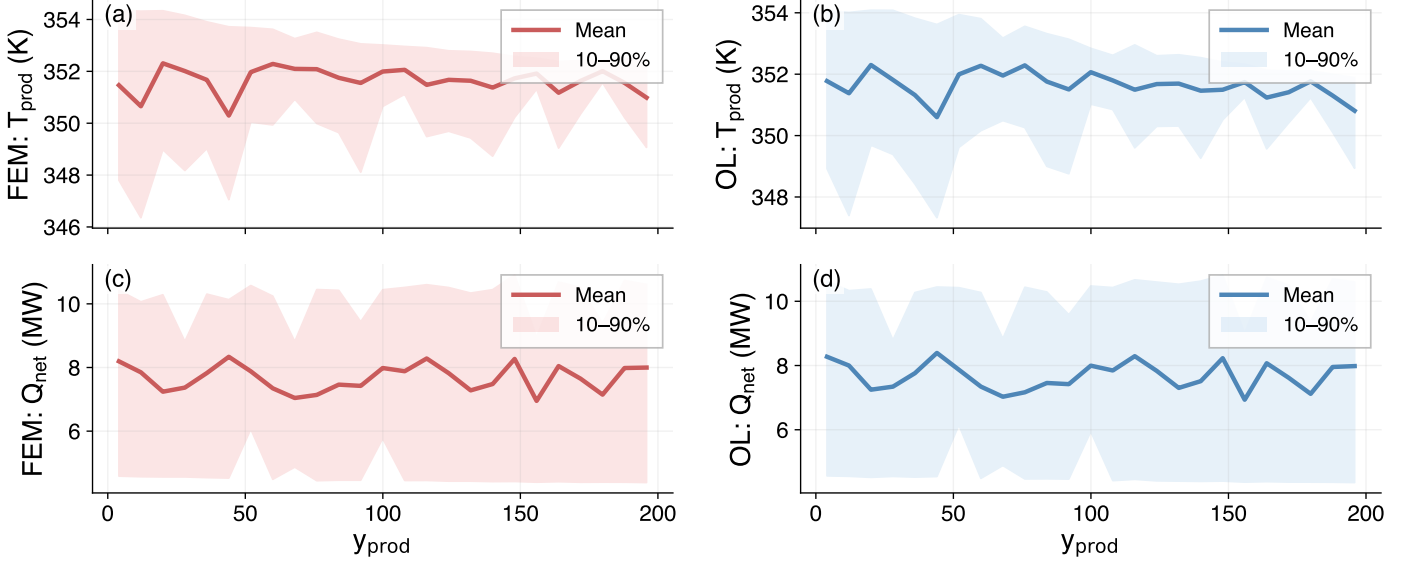


Fig. 5: Univariate sensitivity analysis of the production-well vertical location y_{prod} on (top) the production temperature and (bottom) the net thermal power. The left column shows FEM responses (T_{prod} , Q_{net}), while the right column shows OL surrogate predictions ($T_{\text{prod,OL}}$, $Q_{\text{net,OL}}$). The solid line denotes the mean value, and the shaded band shows the 10th–90th percentiles of the corresponding output computed from all samples that share the same y_{prod} range, while $(\theta_f, s_f, \dot{m}_{\text{inj}})$ vary across the dataset.

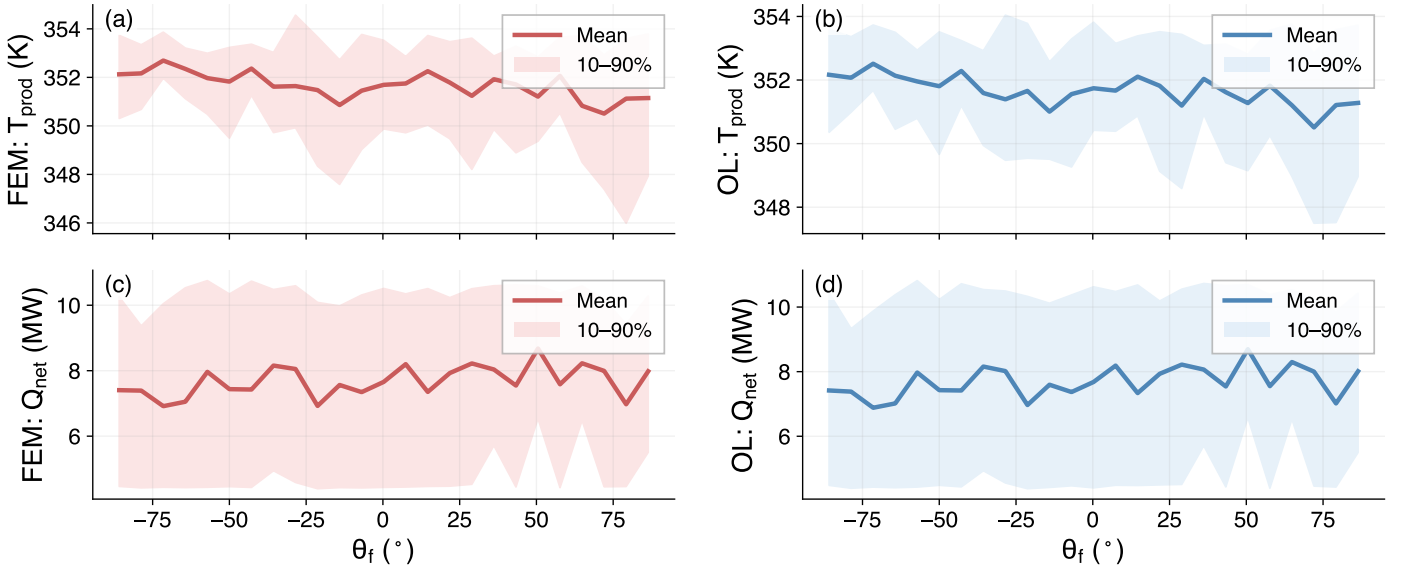


Fig. 6: Univariate sensitivity analysis of the fracture rotation angle θ_f on (top) the production temperature and (bottom) the net thermal power. The left column shows FEM responses (T_{prod} , Q_{net}), while the right column shows OL surrogate predictions ($T_{\text{prod,OL}}$, $Q_{\text{net,OL}}$). The solid line denotes the mean value, and the shaded band shows the 10th–90th percentiles of the corresponding output computed from all samples that share the same θ_f range, while $(y_{\text{prod}}, s_f, \dot{m}_{\text{inj}})$ vary across the dataset.

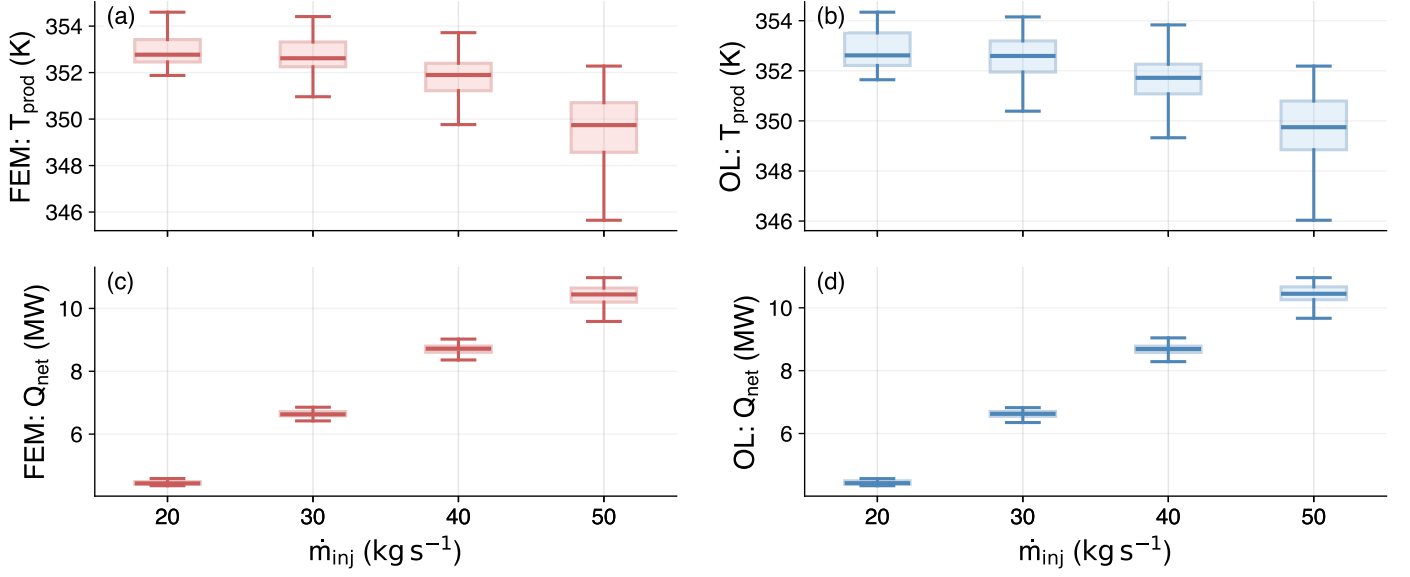


Fig. 7: Univariate sensitivity analysis of the injection mass flow rate $\dot{m}_{inj}$ on (top) the production temperature and (bottom) the net thermal power. The left column shows FEM responses (T_{prod}, Q_{net}), while the right column shows OL surrogate predictions ($T_{prod,OL}, Q_{net,OL}$). For each discrete value of $\dot{m}_{inj}$, boxplots summarize the distribution of the corresponding output across all cases with the same $\dot{m}_{inj}$, while $(y_{prod}, \theta_f, s_f)$ vary across the dataset.

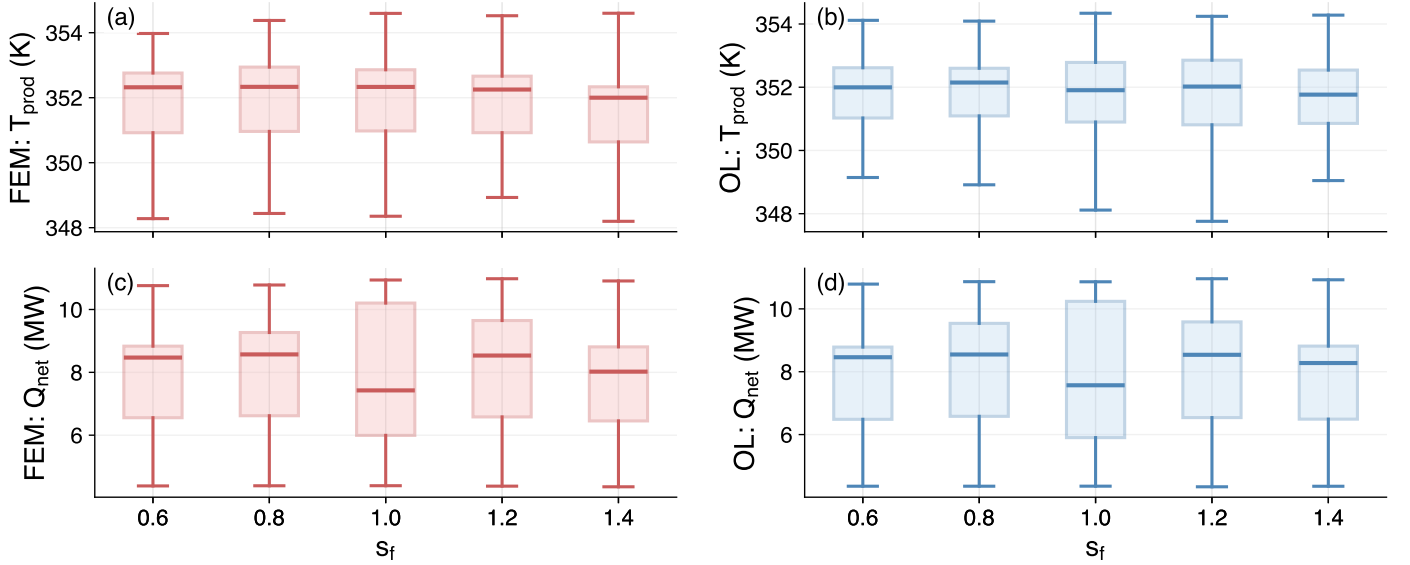


Fig. 8: Univariate sensitivity analysis of the fracture scaling factor s_f on (top) the production temperature and (bottom) the net thermal power. The left column shows FEM responses (T_{prod}, Q_{net}), while the right column shows OL surrogate predictions ($T_{prod,OL}, Q_{net,OL}$). For each discrete value of s_f , boxplots summarize the distribution of the corresponding output across all cases with the same s_f , while $(y_{prod}, \theta_f, \dot{m}_{inj})$ vary across the dataset.

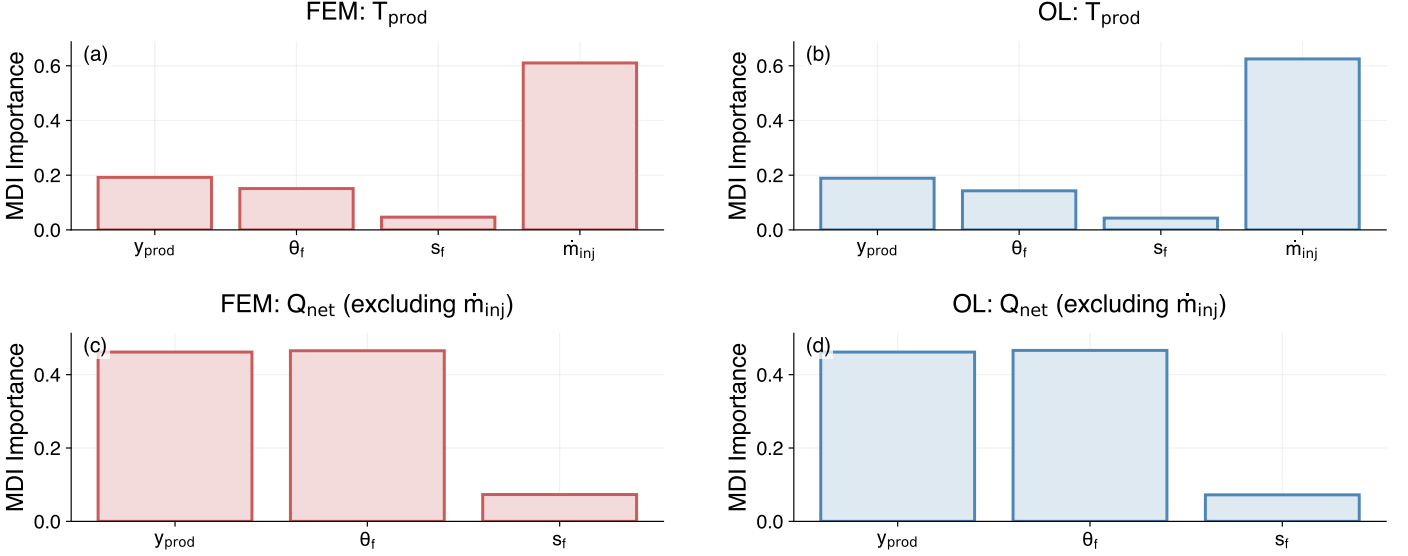


Fig. 9: Feature-importance rankings for the production temperature T_{prod} (top row) and the net thermal power Q_{net} (bottom row), evaluated using the FEM responses (left column) and the operator-learning (OL) surrogate predictions (right column). Panels (a,b) report the relative contributions of $(y_{\text{prod}}, \theta_f, s_f, \dot{m}_{\text{inj}})$ to T_{prod} . Panels (c,d) show the corresponding results for Q_{net} after excluding the injection mass rate $\dot{m}_{\text{inj}}$.

OL results, the analysis is restricted to the test set that is unseen to the OL model. For the net thermal power Q_{net} , an additional importance analysis is reported in which $\dot{m}_{\text{inj}}$ is excluded to avoid dominance arising from the direct proportionality $Q_{\text{net}} \propto \dot{m}_{\text{inj}} (T_{\text{prod}} - T_{\text{inj}})$. The resulting importance rankings for the FEM results are shown in Fig. 9.

Fig. 9 shows that, for both the FEM responses and the OL predictions, the injection mass rate $\dot{m}_{\text{inj}}$ is the dominant feature controlling the production temperature. Fracture-network morphology and well layout also contribute to variations in both T_{prod} and Q_{net} , whereas the fracture scaling factor s_f has a comparatively minor influence. This weak sensitivity to s_f may stem from the simplified, well-connected fracture configurations considered here, for which the hydraulic connectivity remains consistently high across the tested scaling levels.

5 CONCLUSION

Cornell University has supported research on deep geothermal district heating through the Earth Source Heat (ESH) project to support campus decarbonization and its 2035 carbon-neutrality target. Using site constraints from CUBO, this study develops a FEM modeling and reduced order modeling workflow to evaluate the thermo-hydraulic (TH) performance of an injection-production doublet in a fracture-dominated geothermal reservoir.

A fully TH coupled FE model is implemented in MOOSE, with flow and heat transport controlled by a lower-dimensional discrete fracture network embedded in a

porous matrix. The model is benchmarked against Lauwrier’s analytical solution and then used to generate $N = 6000$ FE realizations spanning well placement, operating controls, and fracture-network configurations.

An operator learning surrogate is then trained to learn the nonlinear solution operator mapping fracture geometry, doublet configuration, and injection rate to the temperature field near the production well. On unseen test cases, the surrogate predicts local temperature fields and production temperature with high accuracy and strong generalization (field MSE 0.139 ± 0.011 K; T_{prod} MSE 0.503 ± 0.013 K; $R^2 = 0.892$), while enabling rapid evaluation of the entire test set in ~ 0.1 s on laptop level computing.

Sensitivity analyses and feature importance ranking show that the injection mass rate $\dot{m}_{\text{inj}}$ is the dominant driver of production temperature and strongly affects thermal breakthrough. In contrast, fracture morphology and well spacing exhibit weaker main effects on the mean responses, with their influence emerging primarily through coupled interactions that broaden the response variability across the parameter space. The results further show that the fracture scaling factor s_f has a comparatively minor influence under the present setup, likely because the simplified and consistently well-connected fracture configurations preserve high hydraulic connectivity across the dataset. The OL surrogate reproduces these sensitivity patterns and importance rankings, suggesting that it has learned the underlying FEM response behavior.

The proposed physics-based modeling and operator learning workflow provides Cornell's geothermal development with a fast inference surrogate that learns the FE predicted thermo-hydraulic response of fractured reservoirs. During doublet planning and construction, the surrogate enables real time, high-fidelity predictions for specified well layouts, geological settings, fracture configurations, and operating controls, thereby supporting rapid screening and automatic optimization of well trajectories and spacing during construction, to maximize production temperature and net thermal power under the constraints of high drilling costs. Future work will extend the framework to more complex geology and fracture architectures, incorporate fully coupled thermo-hydro-mechanical processes including stimulation and damage evolution (Liu et al., 2026), further improve model generalizability to previously unseen reservoir scenarios, and incorporate reservoir-scale cooling metrics.

This approach reduces evaluation time by several orders of magnitude relative to finite-element simulation and provides a foundation for real-time reservoir optimization and digital twin development.

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